

IBM DATA ANALYST CAPSTONE PROJECT REPORT

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OUTLINE

- Executive Summary
- Introduction
- Methodology
- Results
 - Visualization – Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion
- Appendices

EXECUTIVE SUMMARY



- **Leading Languages**

- JavaScript is the primary language for both the current year and the language most desired for future use.
- HTML/CSS continue to be significant tools used by developers. Ranking second for the current year and fourth in desired future use.
- SQL, TypeScript, and Python round out the five top languages for both the current year and next year.
- Go and Rust show an increase in desired adoption for the future over current usage.

- **Leading Databases**

- PostgreSQL, MySQL, SQLite, Redis, and MongoDB hold the top five positions for both current year use and future desirability in databases.
- PostgreSQL stands out as not only the most widely used database but also the most desired for future implementation, with over 87.5% more selections than SQLite, which ranks second.

- **Leading Platforms**

- Cloud-based platforms are projected to be the future work environments, with Amazon Web Services(AWS) being the most desired to work with, followed by Microsoft Azure, Google Cloud, and Cloudflare.

INTRODUCTION



- The purpose of this report is to identify current, emerging, and future technical skills used by solution development and data professionals.
- The target audiences include business leaders, policymakers, educators, students, and both current and prospective developers and data professionals.
- The analysis primarily focuses on currently most used programming languages and database systems, as well as those that the global developer community expresses an interest in using in the future.
- Additional analysis was conducted regarding annual compensation in relation to technical skills, programming languages, geographic locations, age, formal education, and experience.

METHODOLOGY

- Data Collection and Sources
 - Retrieved data related to job postings on GitHub, using an API call in Python.
 - Collected public survey data related to Programming Languages and Annual Salary, using web scraping methods in Python.
 - Direct download of the full data set of the Stack Overflow Annual Developer Survey as a CSV file.
- Data Processing using Microsoft Excel and Python to implement standard data cleansing and preparation for analysis.
 - Identified and removed duplicate rows and/or columns
 - Located and managed missing values using an appropriate imputation technique:
 - Multivariate Imputation
 - Regression
 - K-nearest neighbor (KNN)
 - Multiple Imputation by Chained Equations (MICE)
 - Removed rows with missing values where imputation was not appropriate.
 - Identified, evaluated, and handled outliers in the data
 - Identified using Standard Deviation, Interquartile Range (IQR) and Z-Score Methods
 - Handled problematic outliers using an appropriate normalization technique or by dropping the value if appropriate.
- Analysis Tools
 - Microsoft Excel
 - Python using NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Beautiful Soup, Requests, and Miceforest in addition to the Standard Library.
 - IBM Cognos Analytics for Dashboard creation.

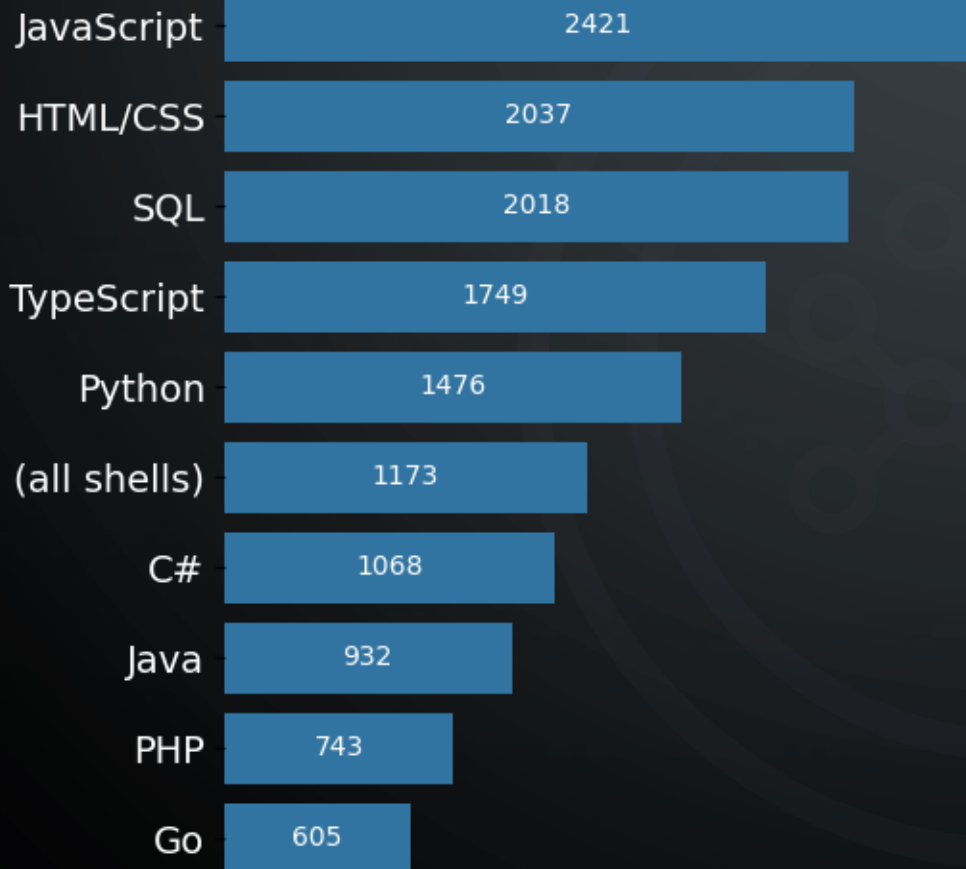


RESULTS

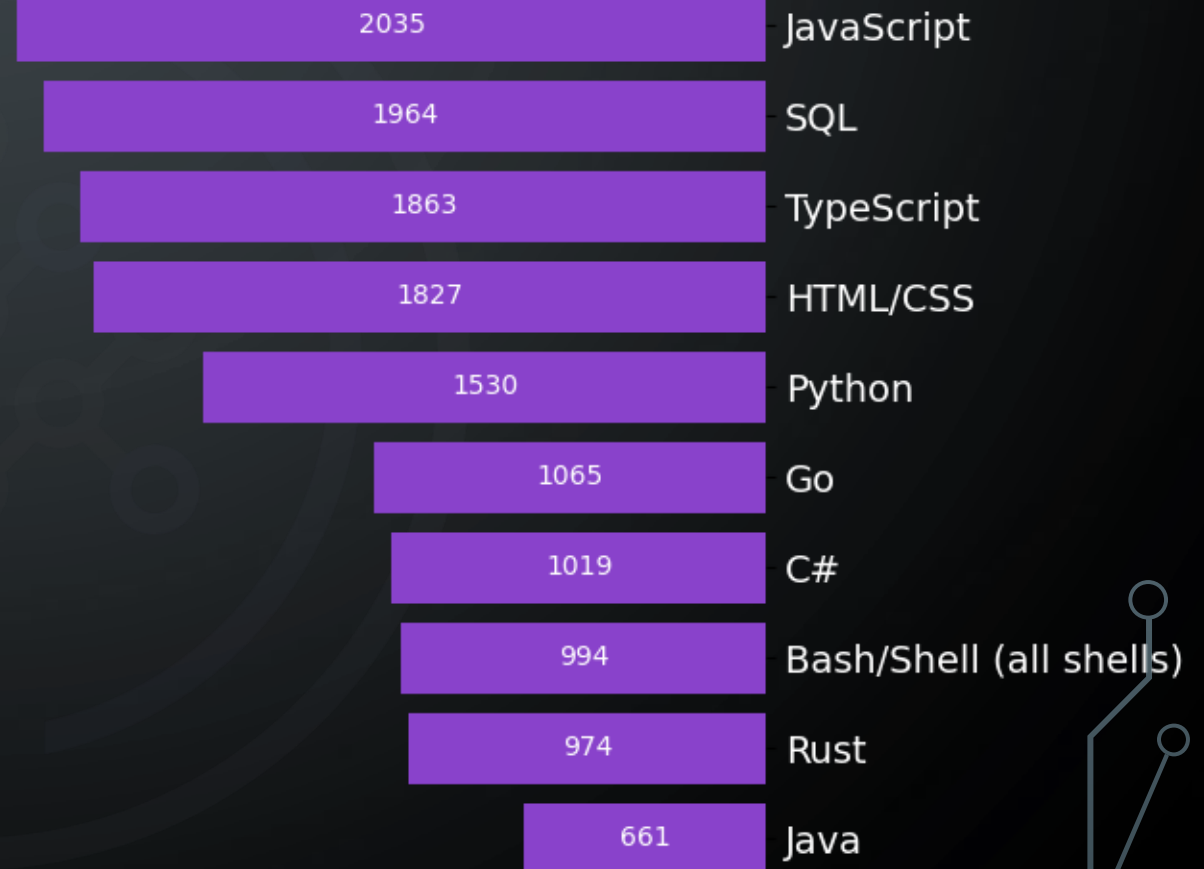


PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS

FINDINGS & IMPLICATIONS

Findings

- JavaScript is currently the most popular language among developers and holds the highest interest for future use.
- HTML/CSS are frequently used skills by developers currently and can be expected to remain desirable in the future.
- Go, TypeScript, Rust, and Python are languages that demonstrate an increased interest for future use higher than their current popularity.
- The C programming language, while not listed among the Top 10 Currently Used or Top 10 Desired skills, is still a highly sought-after skill by employers. (See Appendix A)

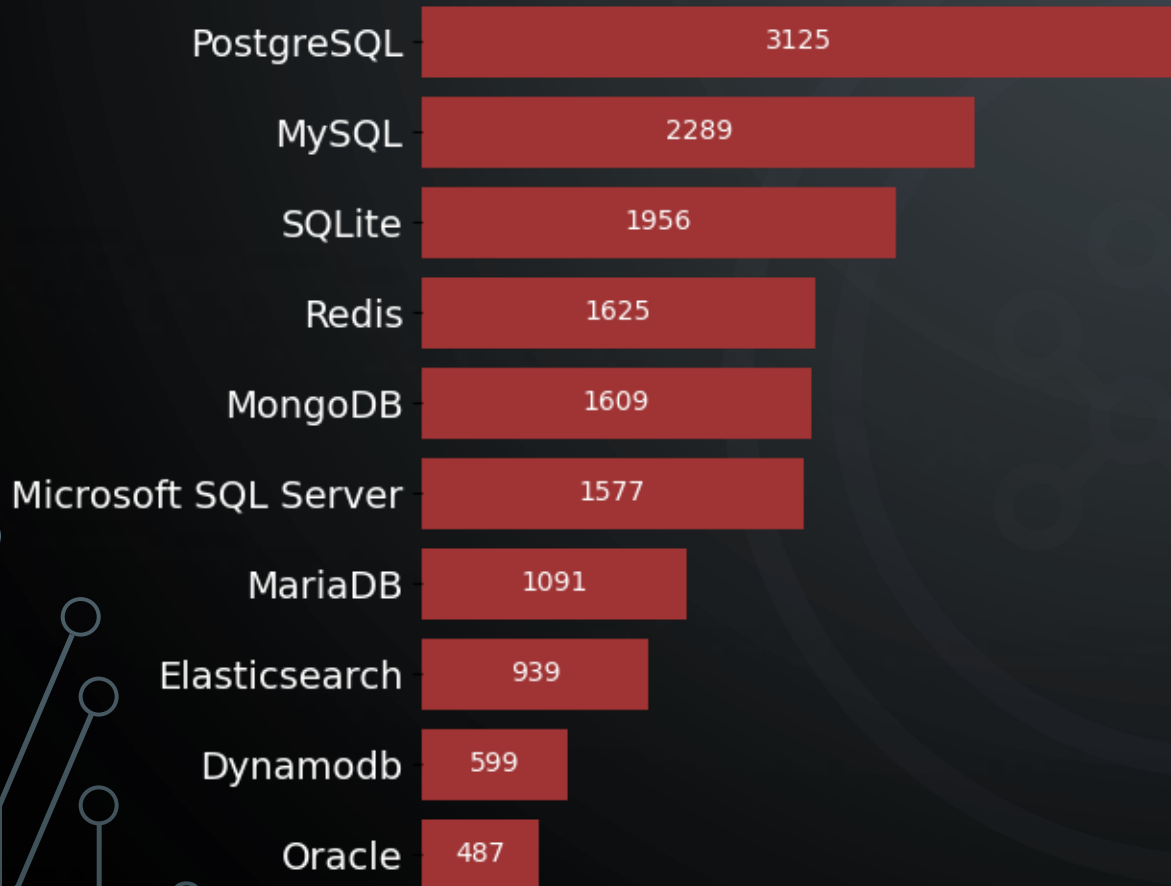
Implications

- The continued popularity of JavaScript, HTML and CSS, plus the increasing popularity of TypeScript implies that web development will remain one of the dominant sectors, for the near future.
- The increasing popularity of both Python and Rust highlights a divided interest between rapid development and the execution speed of solutions. The rise in both may point to advancements in hybrid approaches using both languages.
- Some of the skills employers are looking for, such as C Programming, are not necessarily those that many developers currently use or have an interest in. This may indicate an impending shortage of programmers with the skills to update or maintain legacy solutions. This also provides an opportunity for those willing to develop the needed skills.

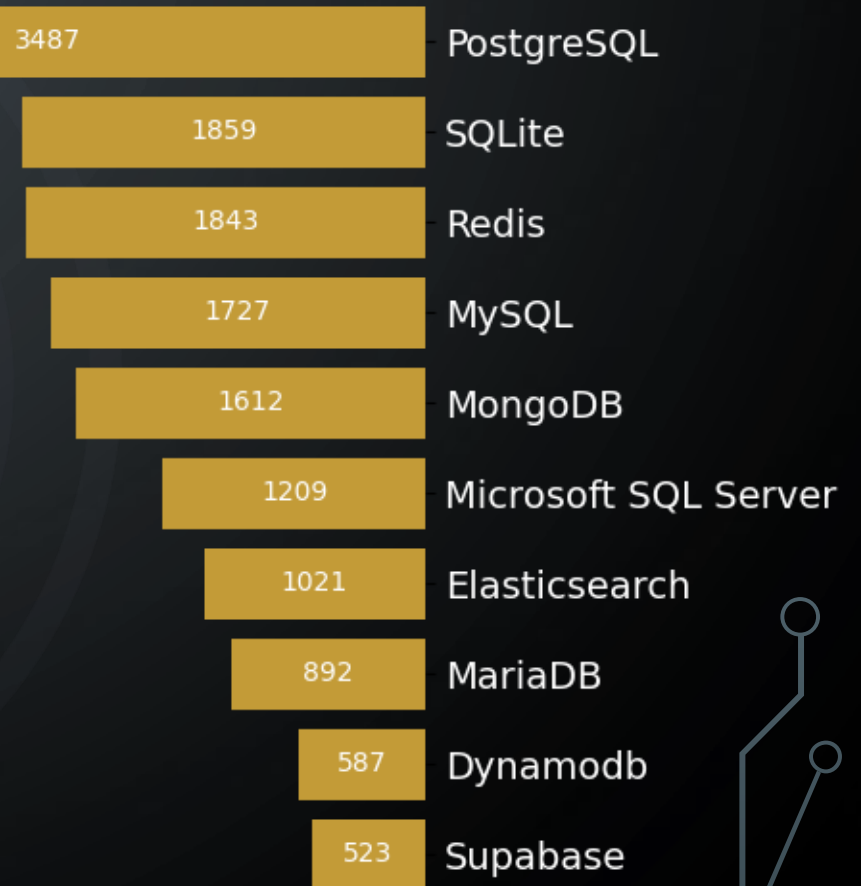


DATABASE TRENDS

Current Year



Next Year



DATABASE TRENDS

FINDINGS & IMPLICATIONS

Findings

- Developer responses show that PostgreSQL was the most used database at the time of the survey. It is also, by far, the one with the greatest expressed interest for future use.
- SQLite, MySQL, and Microsoft SQL Server continue to be widely used and look to remain so for the foreseeable future.
- Redis and MongoDB lead NoSQL database solutions in both current use and future desirability.

Implications

- PostgreSQL is a free and open-source, highly extensible, and secure database that runs on all major operating systems. Its high popularity in the developer community is expected to persist; its continued use or adoption is strongly recommended.
- The increasing interest in Redis, MongoDB, and others demonstrates that the use of NoSQL databases is on the rise. The development of knowledge and skills, as well as the adoption of NoSQL data storage and processes, should be investigated.



DASHBOARD



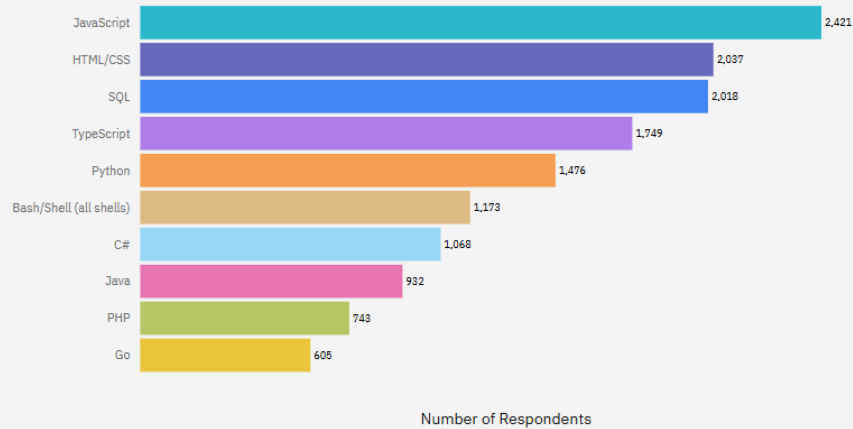
Link to PDF copy IBM Congos Dashboard on Github

https://github.com/Haliaetos/IBM_Data_Analysis/blob/0e7a91d0d37b40aa3af531e61478011b7df9c70e/IBM_Data_Analysis_Capstone_Dashboard.pdf

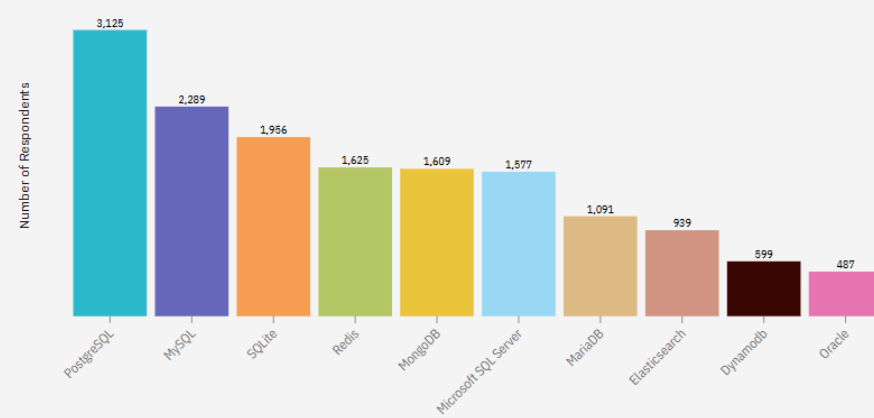


DASHBOARD – CURRENT TECHNOLOGY USAGE

Top 10 Languages Used



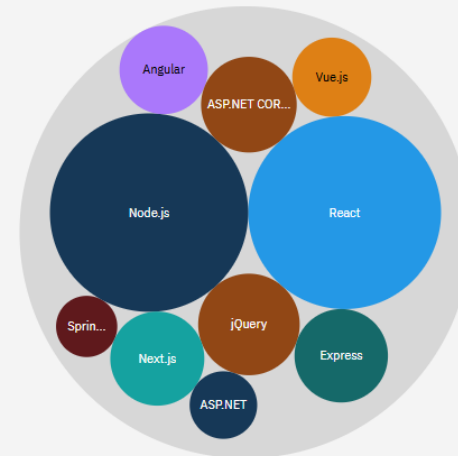
Top 10 Databases Used



Top 10 Platforms Used

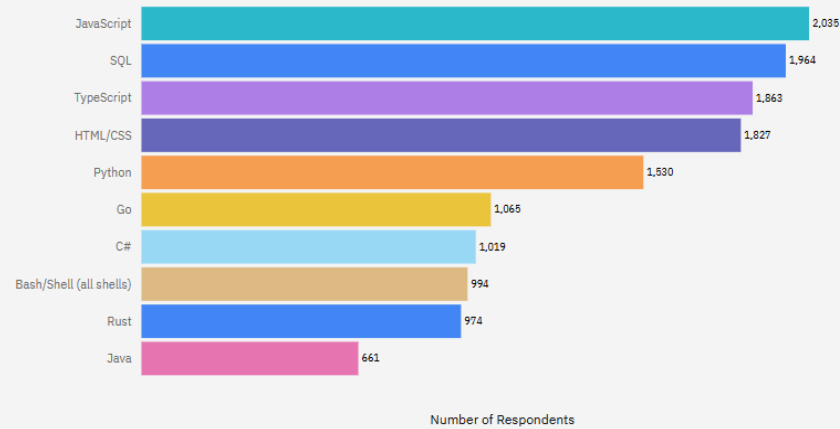


Top 10 Web Frames Used

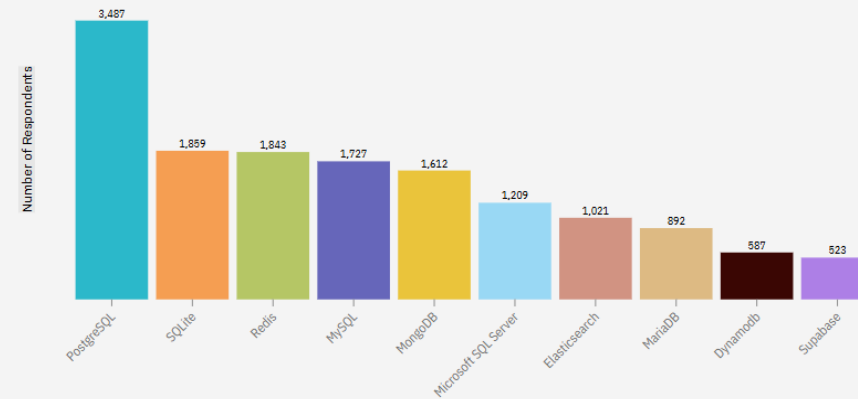


DASHBOARD – FUTURE TECHNOLOGY TRENDS

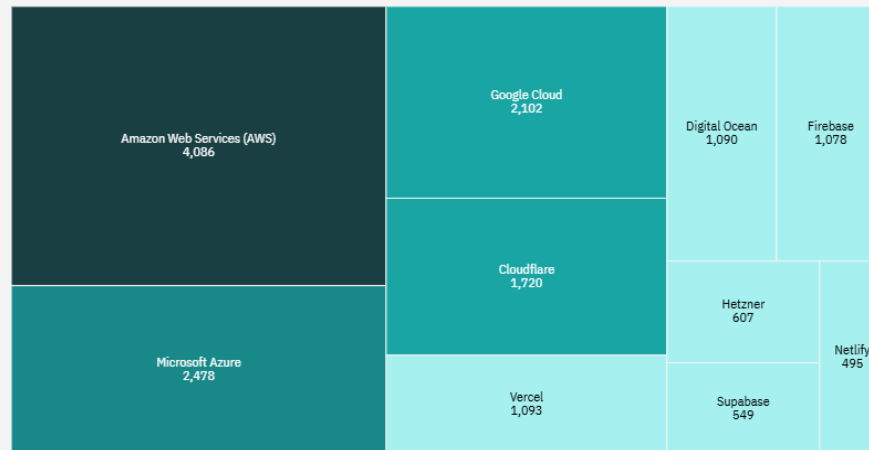
Top 10 Languages Desired Next Year



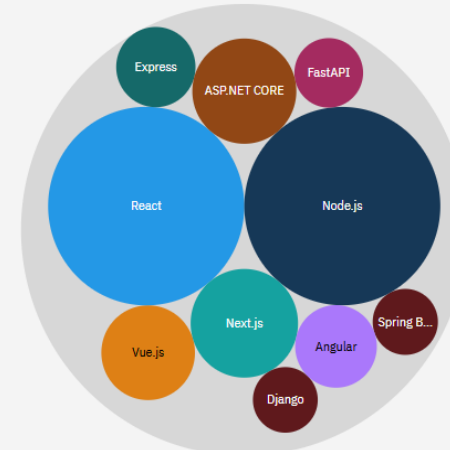
Top 10 Databases Desired Next Year



Top 10 Desired Platforms Desired Next Year

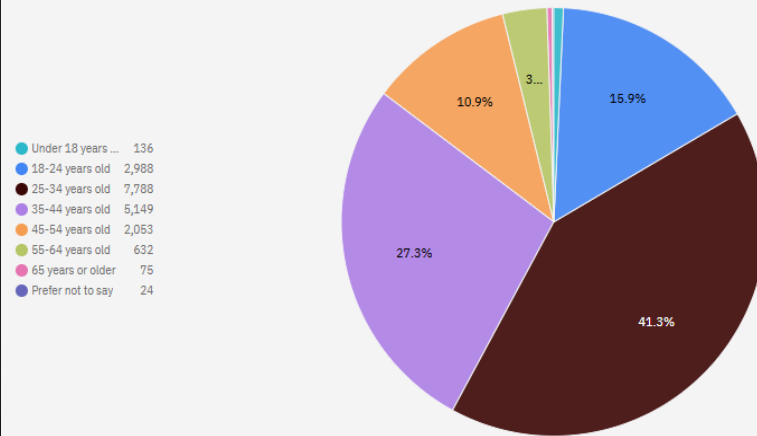


Top 10 Web Frames Desired Next Year

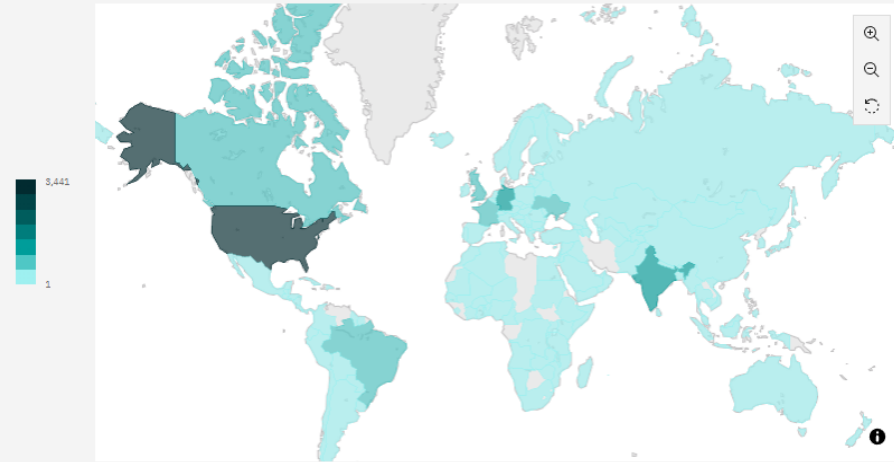


DASHBOARD – DEMOGRAPHICS

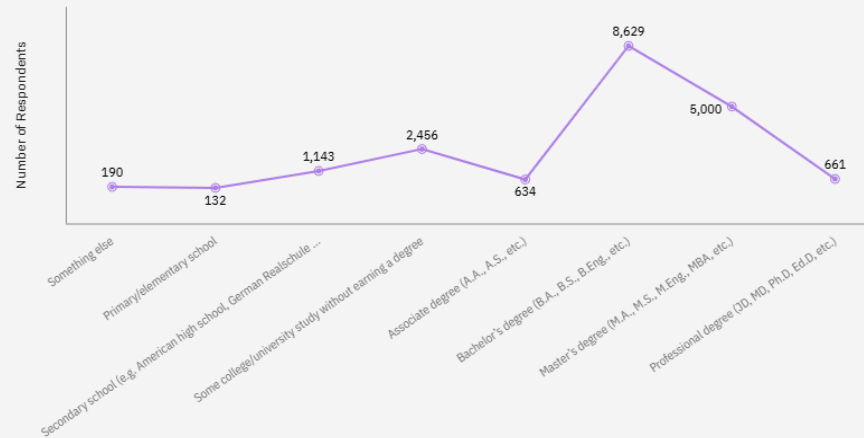
Respondent Distribution by Age



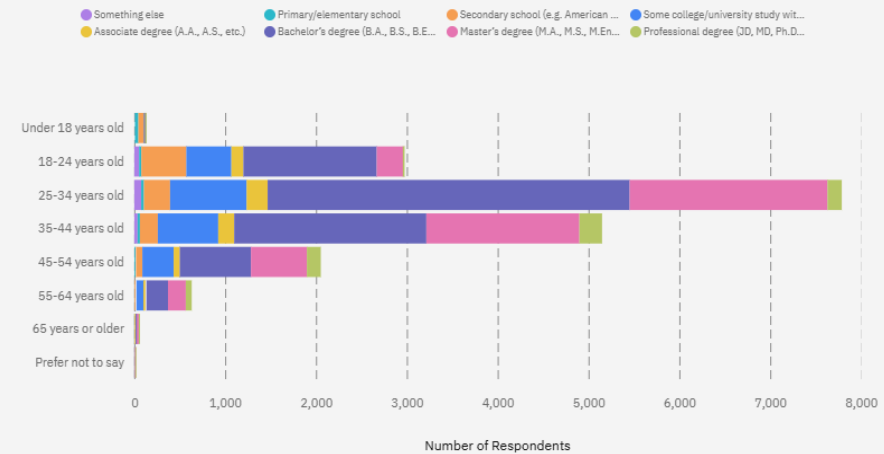
Respondent Count by Country



Respondent Distribution by Formal Education Level



Respondent Count by Age, Classified by Education Level



DISCUSSION



OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript and HTML/CSS were the skills with the highest current use.
- TypeScript is becoming more popular.
- PostgreSQL is currently the most popular database and is gaining further popularity faster than other databases.
- Amazon Web Services(AWS) is the current, and apparent future, front-runner in Cloud Platforms, with Microsoft Azure, Google Cloud, and Cloudflare leading all that follow.
- Node.js and React are currently the most used web frameworks and are poised to continue the trend.
- C Programming was listed as a desired skill in job postings on GitHub 13,498 times, which was five times more than the next most frequently listed skill, Java.
- The vast number of respondents to the Annual Stack Overflow Developer Survey were from the United States, followed by Germany and India.

Implications

- The growing popularity of TypeScript, along with the combined use of Node.js and React, indicates that professionals in the web development field should consider adding these technologies to their toolkit.
- Developers and data specialists should have or develop at least a basic understanding of PostgreSQL, as it continues to be a widely used database solution worldwide. Business leaders and technology decision-makers should explore the benefits of implementing it within their organizations.
- Knowledge of foundational programming languages, especially C, continues to be an essential skill highly sought by employers..



CONCLUSION

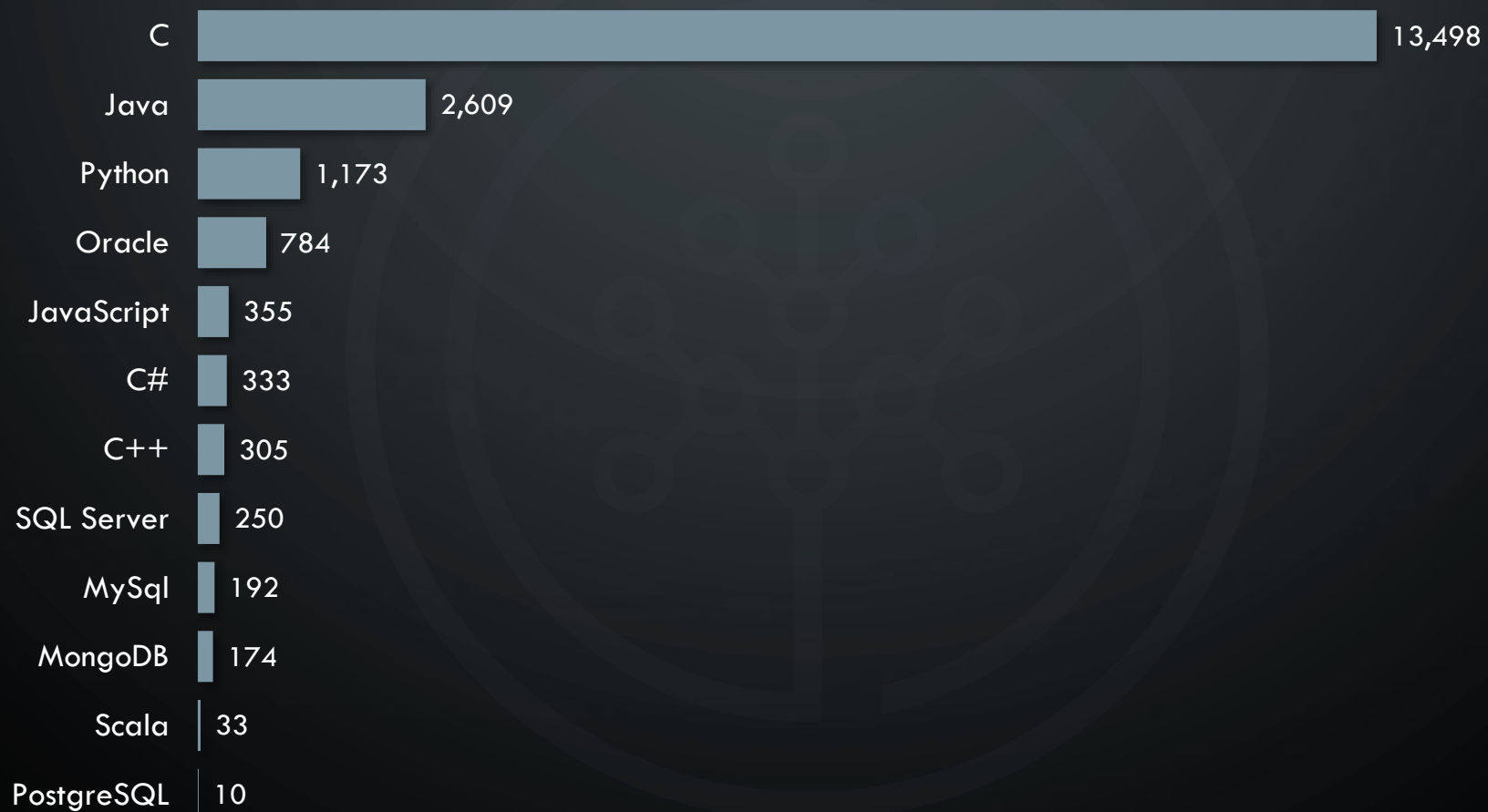
- The most popular skills and tools used this year, and anticipated for next year, are primarily in the web development sector.
- C# and Java are still widely used programming languages, but their popularity may be declining. In contrast, Python, Go, and Rust are gaining increased interest.
- Free and open-source databases, such as PostgreSQL, continue to grow in popularity.
- Cloud-based platforms are becoming more widely used and show increasing popularity, moving forward.
- The high percentage of respondents from the United States might accurately reflect the global distribution of developers. However, it is more likely a result of Stack Overflow's membership dynamics. To gain a better global perspective, it would be beneficial to seek feedback from additional sources in future analyses

APPENDICES



APPENDIX A

JOB POSTINGS BY TECHNOLOGY



APPENDIX B

POPULAR LANGUAGES BY AVERAGE ANNUAL INCOME (USD)

